In [1]:
```
from pandas_datareader import data as wb
import pandas as pd
import numpy as np
import matplotlib.pyplot as pt
from alpha_vantage.timeseries import TimeSeries
import backtrader as btr
import yfinance as yf
import bt
```

In [2]:
```
API_key = "W0TD4SDGAPUFM7QL"

ts = TimeSeries(key=API_key, output_format="pandas")
alpha_vantage = ts.get_daily("AAPL")
apple_stock_data = alpha_vantage[0] #it gives the data from the tup
apple_stock_data = apple_stock_data.iloc[::-1]  # Reverse row order

meta_data = alpha_vantage[1] #it takes the meta data from the tuple

#alpha_vantage[0].to_csv("/Users/ushasthej/Visual_studio/Python for
#alpha_vantage[0].to_excel("/Users/ushasthej/Visual_studio/Python f
#beta = pd.read_csv("/Users/ushasthej/Visual_studio/Python for Fina
#gamma = pd.read_csv("/Users/ushasthej/Visual_studio/Python for Fin
#beta = pd.read_excel("/Users/ushasthej/Visual_studio/Python for Fi
#beta.tail()
```

In [3]:
```
apple_stock_data = yf.download("AAPL", start='2022-11-12')
```

```
[*********************100%***********************]  1 of 1 completed
```

In [4]:
```
apple_stock_data
```

Out[4]:

Price	Close	High	Low	Open	Volume
Ticker	AAPL	AAPL	AAPL	AAPL	AAPL
Date					
2022-11-14	146.575775	148.552789	145.735538	147.257847	73374100
2022-11-15	148.315552	151.824754	146.852566	150.470504	89868300
2022-11-16	147.079880	148.147469	145.597120	147.415983	64218300
2022-11-17	148.987732	149.738992	144.470249	144.747030	80389400
2022-11-18	149.551163	150.944961	148.246342	150.559444	74829600
...	...	...	...	...	...
2025-02-10	227.649994	230.589996	227.199997	229.570007	33115600
2025-02-11	232.619995	235.229996	228.130005	228.199997	53718400
2025-02-12	236.869995	236.960007	230.679993	231.199997	45243300
2025-02-13	241.529999	242.339996	235.570007	236.910004	53614100
2025-02-14	244.600006	245.550003	240.990005	241.250000	40838100

565 rows × 5 columns

In [5]:
```
volume = apple_stock_data[['Volume']]
apple_stock_data['simple_return'] = apple_stock_data['Close'].pct_c
apple_stock_data = apple_stock_data.replace([np.inf, -np.inf], np.n
```

In [6]:
```
window = 20 #number of trading days
apple_stock_data['rolling_volatility'] = apple_stock_data['simple_r
apple_stock_data = apple_stock_data.replace([np.inf, -np.inf], np.n
apple_stock_data
```

Out[6]:

Price	Close	High	Low	Open	Volume	simpl
Ticker	AAPL	AAPL	AAPL	AAPL	AAPL	
Date						
2022-12-13	143.798050	148.246329	142.582191	147.781730	93886200	(
2022-12-14	141.564056	144.974402	139.537615	143.679460	82291200	-(
2022-12-15	134.931152	140.170240	134.466553	139.488168	98931900	-0
2022-12-16	132.964035	136.067946	132.193001	135.118988	160156900	-(
2022-12-19	130.848633	133.646109	129.810713	133.557147	79592600	-
...	...	...	...	...	...	
2025-02-10	227.649994	230.589996	227.199997	229.570007	33115600	
2025-02-11	232.619995	235.229996	228.130005	228.199997	53718400	(
2025-02-12	236.869995	236.960007	230.679993	231.199997	45243300	(
2025-02-13	241.529999	242.339996	235.570007	236.910004	53614100	(
2025-02-14	244.600006	245.550003	240.990005	241.250000	40838100	

545 rows × 7 columns

In [7]:

```
data = apple_stock_data.copy()
def avg_true_range(datas, win = 14):
    high_low = datas['High'] - datas['Low']
    high_close = abs(datas['High'] - datas['Close'].shift())
    low_close = abs(datas['Low'] - datas['Close'].shift())
    true_range = pd.concat([high_low, high_close, low_close], axis
    return true_range.rolling(window=win).mean()

apple_stock_data['avg_true_value'] = avg_true_range(data)
apple_stock_data = apple_stock_data.replace([np.inf, -np.inf], np.n
apple_stock_data
```

Out[7]:

Price	Close	High	Low	Open	Volume	simple
Ticker	AAPL	AAPL	AAPL	AAPL	AAPL	
Date						
2023-01-03	123.632515	129.395502	122.742858	128.782633	112117500	-0
2023-01-04	124.907707	127.181276	123.642420	125.431615	89113600	0
2023-01-05	123.583099	126.301493	123.326093	125.668849	80962700	-0
2023-01-06	128.130234	128.792531	123.454601	124.561732	87754700	0
2023-01-09	128.654144	131.876686	128.397138	128.970474	70790800	0.
...	...	...	...	...	...	
2025-02-10	227.649994	230.589996	227.199997	229.570007	33115600	0
2025-02-11	232.619995	235.229996	228.130005	228.199997	53718400	0
2025-02-12	236.869995	236.960007	230.679993	231.199997	45243300	0
2025-02-13	241.529999	242.339996	235.570007	236.910004	53614100	0
2025-02-14	244.600006	245.550003	240.990005	241.250000	40838100	

532 rows × 8 columns

In [8]:

```
atr_data = apple_stock_data[['Close', 'simple_return', 'rolling_vol
atr_data['signal'] = np.zeros(len(atr_data['Close']))
atr_data
```

```
/var/folders/98/n7cbxwmj7z1cdgwpqvk8njpr0000gn/T/ipykernel_16816/180
8871403.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pand
as-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-co
py
  atr_data['signal'] = np.zeros(len(atr_data['Close']))
```

Out[8]:

Price	Close	simple_return	rolling_volatility	avg_true_value	signal
Ticker	AAPL				
Date					
2023-01-03	123.632515	-0.037405	0.019779	4.518885	0.0
2023-01-04	124.907707	0.010314	0.020205	4.367786	0.0
2023-01-05	123.583099	-0.010605	0.019772	4.191973	0.0
2023-01-06	128.130234	0.036794	0.021907	4.066289	0.0
2023-01-09	128.654144	0.004089	0.021669	4.057111	0.0
...	...	...	...	...	...
2025-02-10	227.649994	0.001187	0.020328	6.490988	0.0
2025-02-11	232.619995	0.021832	0.020876	6.723470	0.0
2025-02-12	236.869995	0.018270	0.021239	6.834557	0.0
2025-02-13	241.529999	0.019673	0.021238	7.017030	0.0
2025-02-14	244.600006	0.012711	0.018986	6.674194	0.0

532 rows × 5 columns

In [9]:

```
datas = atr_data.copy()
# Defining thresholds
# Compute rolling metrics
volume_threshold_data = datas['rolling_volatility'].rolling(window=
rolling_mean_10 = datas['Close'].rolling(window=10).mean()
rolling_max_10 = datas['Close'].rolling(window=10).max()

# Remove first 9 rows (NaN values from rolling)
datas = datas.iloc[9:]  # Copy to avoid modifying the original Data
rolling_mean_10 = rolling_mean_10.iloc[9:]
rolling_max_10 = rolling_max_10.iloc[9:]
volume_threshold_data = volume_threshold_data.iloc[9:]

# Ensure the signal column exists
if 'signal' not in datas.columns:
    datas['signal'] = np.nan

for i in range(datas.shape[0]):
    # Extract single values (scalars)
```

```
    rolling_mean_value = rolling_mean_10.iloc[i].item()  # No .item
    rolling_max_value = rolling_max_10.iloc[i].item()
    volume_threshold = volume_threshold_data.iloc[i].item()

    # Short signal
    if datas['rolling_volatility'].iloc[i].item() < volume_threshol
        if datas['Close'].iloc[i].item() < rolling_mean_value:
            datas.iloc[i, datas.columns.get_loc('signal')] = -1

    # Long signal
    else:
        if datas['Close'].iloc[i].item() >= rolling_max_value:
            datas.iloc[i, datas.columns.get_loc('signal')] = 1



atr_signal = datas.copy()

atr_signal#['signal'].value_counts()
```

Out[9]:

Price	Close	simple_return	rolling_volatility	avg_true_value	signal
Ticker	AAPL				
Date					
2023-01-17	134.377594	0.008756	0.019442	3.757028	0.0
2023-01-18	133.656006	-0.005370	0.019187	3.788099	0.0
2023-01-19	133.715302	0.000444	0.018798	3.598870	0.0
2023-01-20	136.285431	0.019221	0.019214	3.553680	0.0
2023-01-23	139.488205	0.023500	0.019195	3.760562	0.0
...	...	...	...	...	...
2025-02-10	227.649994	0.001187	0.020328	6.490988	0.0
2025-02-11	232.619995	0.021832	0.020876	6.723470	0.0
2025-02-12	236.869995	0.018270	0.021239	6.834557	0.0
2025-02-13	241.529999	0.019673	0.021238	7.017030	1.0
2025-02-14	244.600006	0.012711	0.018986	6.674194	0.0

523 rows × 5 columns

backtest 1

In [10]:
```
def backtest_strategy(datas2: pd.DataFrame):
    #calculate position and strategy returns, tarding the next day
    datas2['simple_returns1'] = (datas2['Close']/datas2['Close'].sh
    datas2['position'] = datas2['signal'].ffill()
    datas2['strategy_returns'] = datas2['position'].shift(1) * data

    return datas2

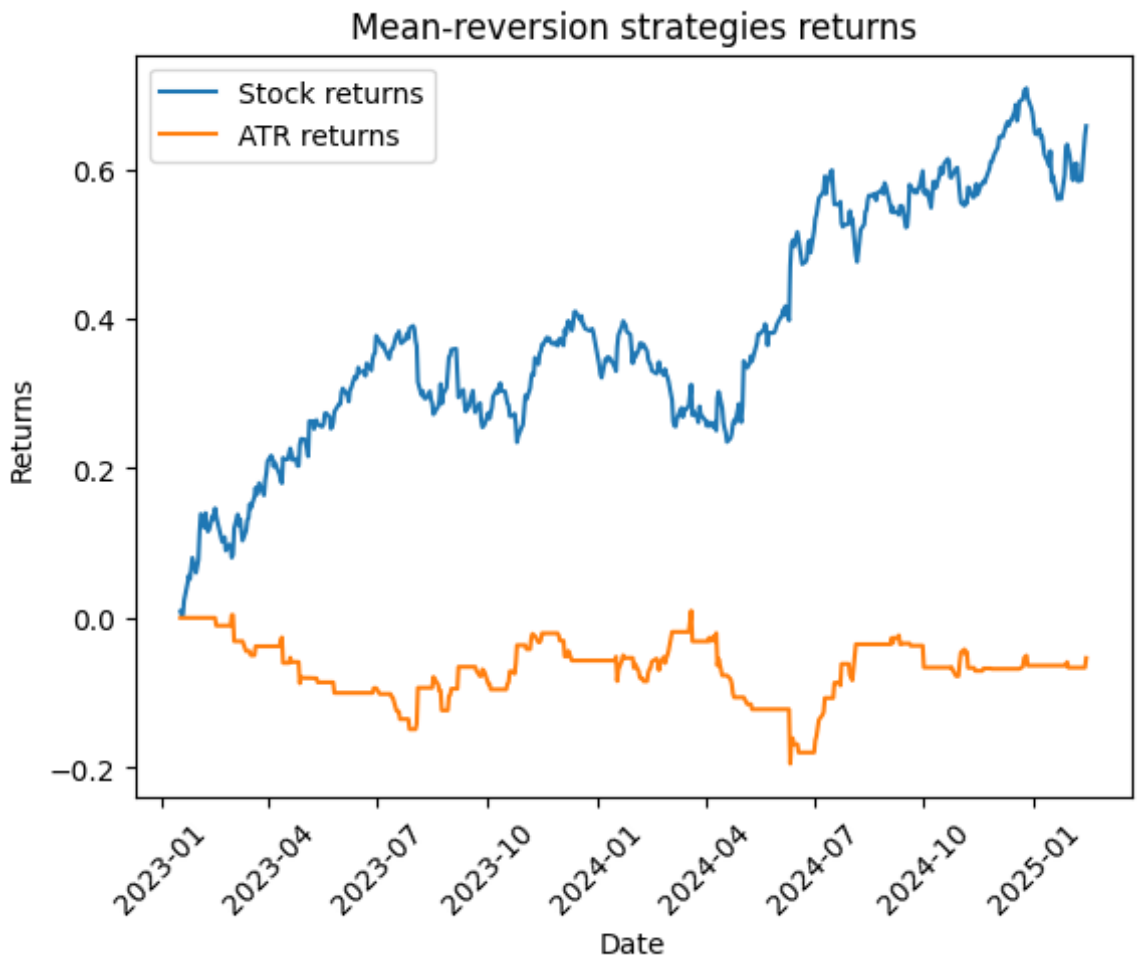
atr_strategy = backtest_strategy(atr_signal).fillna(0)
atr_strategy['strategy_returns'].value_counts()
```

Out[10]:
```
strategy_returns
 0.000000    368
-0.006890      1
 0.008627      1
-0.021863      1
-0.005091      1
            ...
-0.005181      1
 0.014451      1
 0.005885      1
-0.002624      1
 0.012711      1
Name: count, Length: 156, dtype: int64
```

In [11]:
```
print('-----------------------------------')
print("RSI strategy returns:", atr_strategy['strategy_returns'].cum
print("Stock returns:", atr_strategy['simple_return'].cumsum().iloc
print('-----------------------------------')

#pt.figure(figsize=(12,5))
pt.plot(atr_strategy.index, atr_strategy['simple_return'].cumsum(),
pt.plot(atr_strategy.index, atr_strategy['strategy_returns'].cumsum
pt.title('Mean-reversion strategies returns')
pt.xlabel('Date')
pt.ylabel('Returns')
pt.xticks(rotation=45)
pt.legend()
pt.show()
```

```
-----------------------------------
RSI strategy returns: -0.053805480112578796
Stock returns: 0.6575966280532961
-----------------------------------
```



In [12]:
```
def backtest_strategy_alt(data: pd.DataFrame):
    # Ensure index is in datetime format
    data.index = pd.to_datetime(data.index)

    position = False
    price_change = []

    for i in data.index:
        if i not in data.index:
            print(f"Skipping missing date: {i}")
            continue  # Skip missing dates

        price = data.loc[i, "Close"].item()
        date = i
        i_pos = data.index.get_loc(i)  # Get integer position of i

        # Long position
        if (data.loc[i, "signal"].item() == 1) and (not position):
            buy_price = price
            position = True
            data.at[i, 'buy_date'] = date
            print(f"Long position open at {buy_price} on {date}")

            #close position after 15% profit or 3% loss or close im
            for k in range(i_pos + 1, min(i_pos + 46, len(data))):
                market_price = data["Close"].iloc[k].item()
                percent = (market_price/price) - 1
                if k < (i_pos + 46):
```

```
                    if (percent >= 0.05) and position:
                        price_change.append(percent)
                        position = False
                        print(f"Long position closed at {market_pri
                        break

                    elif (percent <= -0.005) and position:
                        price_change.append(percent)
                        position = False
                        print(f"Long position closed at {market_pri
                        break

                else:
                    price_change.append(percent)
                    position = False
                    print(f"Long position closed at {market_price}
                    break


        # Short position
        elif (data.loc[i, "signal"].item() == -1) and (not position
            sell_price = price
            position = True
            data.at[i, 'sell_date'] = date
            print(f"Short position open at {sell_price} on {date}")

            #close position after 5% profit or 1% loss or close imm
            for k in range(i_pos + 1, min(i_pos + 46, len(data))):
                market_price = data["Close"].iloc[k].item()
                percent = 1 - (market_price/price)
                if k < (i_pos + 46):
                    if (percent >= 0.05) and position:
                        price_change.append(percent)
                        position = False
                        print(f"Short position closed at {market_pr
                        break

                    elif (percent <= -0.005) and position == True:
                        price_change.append(percent)
                        position = False
                        print(f"Short position closed at {market_pr
                        break

                else:
                    price_change.append(percent)
                    position = False
                    print(f"Short position closed at {market_price}
                    break

    trades_count = len(price_change)
    won_trades = sum(1 for x in price_change if x > 0)
    lost_trades = sum(1 for x in price_change if x < 0)

    # Recursively calculate return
    def calculate_return(total_return, price_change, i):
        if i == len(price_change):
```

```
            return total_return
        return calculate_return(total_return * (price_change[i] + 1

    total_return = round((calculate_return(1, price_change, 0) - 1)

    print("\n--------------------------------")
    print(f"Number of Trades: {trades_count}")
    print(f"Wins: {won_trades}")
    print(f"Losses: {lost_trades}")
    print(f"Total Returns: {total_return}%\n")

# Run backtest with atr_signal DataFrame
backtest_strategy_alt(atr_signal)

signal_df = atr_signal.copy()
pt.figure(figsize=(12,5))
pt.xticks(rotation=45)

pt.plot(signal_df.index, signal_df['Close'])
pt.scatter(signal_df[signal_df['signal'] == 1].index, signal_df[sig
pt.scatter(signal_df[signal_df['signal'] == -1].index, signal_df[si

pt.title('Volatility strategy', fontweight="bold")
pt.legend()
pt.show()
```

```
Long position open at 153.77919006347656 on 2023-02-15 00:00:00
Long position closed at 152.17535400390625 on 2023-02-16 00:00:00
Short position open at 145.93826293945312 on 2023-02-28 00:00:00
Short position closed at 149.52212524414062 on 2023-03-03 00:00:00
Short position open at 144.45321655273438 on 2023-03-02 00:00:00
Short position closed at 149.52212524414062 on 2023-03-03 00:00:00
Short position open at 147.01736450195312 on 2023-03-10 00:00:00
Short position closed at 148.96771240234375 on 2023-03-13 00:00:00
Long position open at 154.29400634765625 on 2023-03-16 00:00:00
Long position closed at 153.45245361328125 on 2023-03-17 00:00:00
Long position open at 155.82852172851562 on 2023-03-20 00:00:00
Long position closed at 164.5109405517578 on 2023-04-03 00:00:00
Short position open at 160.4122772216797 on 2023-04-10 00:00:00
Short position closed at 163.9070587158203 on 2023-04-13 00:00:00
Short position open at 159.19456481933594 on 2023-04-11 00:00:00
Short position closed at 163.9070587158203 on 2023-04-13 00:00:00
Short position open at 158.50155639648438 on 2023-04-12 00:00:00
Short position closed at 163.9070587158203 on 2023-04-13 00:00:00
Long position open at 164.80795288085938 on 2023-04-18 00:00:00
Long position closed at 163.37245178222656 on 2023-04-21 00:00:00
Long position open at 165.9563751220703 on 2023-04-19 00:00:00
Long position closed at 164.9861602783203 on 2023-04-20 00:00:00
Short position open at 162.13491821289062 on 2023-04-25 00:00:00
Short position closed at 166.7285919189453 on 2023-04-27 00:00:00
Short position open at 162.125 on 2023-04-26 00:00:00
Short position closed at 166.7285919189453 on 2023-04-27 00:00:00
Long position open at 166.7285919189453 on 2023-04-27 00:00:00
Long position closed at 165.7781524658203 on 2023-05-03 00:00:00
Long position open at 167.98590087890625 on 2023-04-28 00:00:00
Long position closed at 166.85726928710938 on 2023-05-02 00:00:00
```

```
Long position open at 171.8370819091797 on 2023-05-05 00:00:00
Long position closed at 170.05503845214844 on 2023-05-09 00:00:00
Long position open at 172.0152587890625 on 2023-05-11 00:00:00
Long position closed at 171.08335876464844 on 2023-05-12 00:00:00
Short position open at 171.49977111816406 on 2023-05-25 00:00:00
Short position closed at 173.918701171875 on 2023-05-26 00:00:00
Long position open at 186.43991088867188 on 2023-06-27 00:00:00
Long position closed at 180.4222412109375 on 2023-08-04 00:00:00
Long position open at 192.2990264892578 on 2023-06-30 00:00:00
Long position closed at 190.80201721191406 on 2023-07-03 00:00:00
Short position open at 188.1352081298828 on 2023-07-12 00:00:00
Short position closed at 192.31883239746094 on 2023-07-17 00:00:00
Short position open at 188.8985595703125 on 2023-07-13 00:00:00
Short position closed at 192.31883239746094 on 2023-07-17 00:00:00
Short position open at 189.0472869873047 on 2023-07-14 00:00:00
Short position closed at 192.31883239746094 on 2023-07-17 00:00:00
Long position open at 192.31883239746094 on 2023-07-17 00:00:00
Long position closed at 190.2864990234375 on 2023-07-21 00:00:00
Long position open at 193.41929626464844 on 2023-07-19 00:00:00
Long position closed at 191.4662628173828 on 2023-07-20 00:00:00
Short position open at 191.55548095703125 on 2023-07-27 00:00:00
Short position closed at 194.14300537109375 on 2023-07-28 00:00:00
Short position open at 190.92098999023438 on 2023-08-02 00:00:00
Short position closed at 180.4222412109375 on 2023-08-04 00:00:00
Short position open at 189.5231170654297 on 2023-08-03 00:00:00
Short position closed at 177.30926513671875 on 2023-08-07 00:00:00
Short position open at 175.28529357910156 on 2023-08-16 00:00:00
Short position closed at 179.80218505859375 on 2023-08-23 00:00:00
Short position open at 172.73399353027344 on 2023-08-17 00:00:00
Short position closed at 174.5605926513672 on 2023-08-21 00:00:00
Short position open at 173.22044372558594 on 2023-08-18 00:00:00
Short position closed at 174.5605926513672 on 2023-08-21 00:00:00
Short position open at 174.5605926513672 on 2023-08-21 00:00:00
Short position closed at 175.9404754638672 on 2023-08-22 00:00:00
Long position open at 179.80218505859375 on 2023-08-23 00:00:00
Long position closed at 175.09666442871094 on 2023-08-24 00:00:00
Long position open at 182.78033447265625 on 2023-08-29 00:00:00
Long position closed at 181.5791778564453 on 2023-09-06 00:00:00
Long position open at 186.28465270996094 on 2023-08-30 00:00:00
Long position closed at 181.5791778564453 on 2023-09-06 00:00:00
Long position open at 186.50306701660156 on 2023-08-31 00:00:00
Long position closed at 181.5791778564453 on 2023-09-06 00:00:00
Short position open at 181.5791778564453 on 2023-09-06 00:00:00
Short position closed at 170.70883178710938 on 2023-09-26 00:00:00
Short position open at 172.6644744873047 on 2023-09-21 00:00:00
Short position closed at 174.7988739013672 on 2023-09-25 00:00:00
Short position open at 173.5182342529297 on 2023-09-22 00:00:00
Short position closed at 174.7988739013672 on 2023-09-25 00:00:00
Short position open at 170.70883178710938 on 2023-09-26 00:00:00
Short position closed at 172.4857940673828 on 2023-10-02 00:00:00
Short position open at 169.18995666503906 on 2023-09-27 00:00:00
Short position closed at 172.4857940673828 on 2023-10-02 00:00:00
Short position open at 169.4480743408203 on 2023-09-28 00:00:00
Short position closed at 172.4857940673828 on 2023-10-02 00:00:00
Short position open at 169.96429443359375 on 2023-09-29 00:00:00
Short position closed at 172.4857940673828 on 2023-10-02 00:00:00
```

```
Short position open at 171.14561462402344 on 2023-10-03 00:00:00
Short position closed at 172.39646911621094 on 2023-10-04 00:00:00
Short position open at 175.86105346679688 on 2023-10-17 00:00:00
Short position closed at 165.67572021484375 on 2023-10-26 00:00:00
Short position open at 174.5605926513672 on 2023-10-18 00:00:00
Short position closed at 165.67572021484375 on 2023-10-26 00:00:00
Short position open at 174.18336486816406 on 2023-10-19 00:00:00
Short position closed at 176.27801513671875 on 2023-11-02 00:00:00
Short position open at 171.6221466064453 on 2023-10-20 00:00:00
Short position closed at 172.7042236328125 on 2023-11-01 00:00:00
Short position open at 171.7412567138672 on 2023-10-23 00:00:00
Short position closed at 172.7042236328125 on 2023-11-01 00:00:00
Short position open at 172.17807006835938 on 2023-10-24 00:00:00
Short position closed at 176.27801513671875 on 2023-11-02 00:00:00
Short position open at 169.8551025390625 on 2023-10-25 00:00:00
Short position closed at 172.7042236328125 on 2023-11-01 00:00:00
Long position open at 176.27801513671875 on 2023-11-02 00:00:00
Long position closed at 175.36471557617188 on 2023-11-03 00:00:00
Long position open at 177.92591857910156 on 2023-11-06 00:00:00
Long position closed at 186.88795471191406 on 2023-11-15 00:00:00
Long position open at 180.49710083007812 on 2023-11-07 00:00:00
Long position closed at 190.30740356445312 on 2023-11-20 00:00:00
Long position open at 181.55931091308594 on 2023-11-08 00:00:00
Long position closed at 192.26565551757812 on 2023-12-05 00:00:00
Long position open at 185.2875518798828 on 2023-11-10 00:00:00
Long position closed at 183.6970977783203 on 2023-11-13 00:00:00
Long position open at 186.32135009765625 on 2023-11-14 00:00:00
Long position closed at 196.77857971191406 on 2023-12-13 00:00:00
Long position open at 186.88795471191406 on 2023-11-15 00:00:00
Long position closed at 196.77857971191406 on 2023-12-13 00:00:00
Long position open at 188.5778045654297 on 2023-11-16 00:00:00
Long position closed at 184.53208923339844 on 2024-01-02 00:00:00
Short position open at 188.2398223876953 on 2023-11-29 00:00:00
Short position closed at 190.09866333007812 on 2023-12-01 00:00:00
Short position open at 188.81637573242188 on 2023-11-30 00:00:00
Short position closed at 190.09866333007812 on 2023-12-01 00:00:00
Short position open at 188.29945373535156 on 2023-12-04 00:00:00
Short position closed at 192.26565551757812 on 2023-12-05 00:00:00
Long position open at 193.11058044433594 on 2023-12-07 00:00:00
Long position closed at 192.02706909179688 on 2023-12-11 00:00:00
Long position open at 194.5419921875 on 2023-12-08 00:00:00
Long position closed at 192.02706909179688 on 2023-12-11 00:00:00
Short position open at 182.53408813476562 on 2024-01-16 00:00:00
Short position closed at 187.50424194335938 on 2024-01-18 00:00:00
Short position open at 181.58973693847656 on 2024-01-17 00:00:00
Short position closed at 187.50424194335938 on 2024-01-18 00:00:00
Long position open at 187.50424194335938 on 2024-01-18 00:00:00
Long position closed at 183.29949951171875 on 2024-01-31 00:00:00
Long position open at 190.416748046875 on 2024-01-19 00:00:00
Long position closed at 186.91775512695312 on 2024-01-30 00:00:00
Long position open at 192.7328338623047 on 2024-01-22 00:00:00
Long position closed at 191.27162170410156 on 2024-01-26 00:00:00
Long position open at 194.01515197753906 on 2024-01-23 00:00:00
Long position closed at 193.01116943359375 on 2024-01-25 00:00:00
Short position open at 183.29949951171875 on 2024-01-31 00:00:00
Short position closed at 185.74481201171875 on 2024-02-01 00:00:00
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Short position open at 185.74481201171875 on 2024-02-01 00:00:00
Short position closed at 188.1702423095703 on 2024-02-06 00:00:00
Short position open at 184.7408447265625 on 2024-02-02 00:00:00
Short position closed at 186.55992126464844 on 2024-02-05 00:00:00
Short position open at 186.55992126464844 on 2024-02-05 00:00:00
Short position closed at 188.1702423095703 on 2024-02-06 00:00:00
Short position open at 188.1702423095703 on 2024-02-06 00:00:00
Short position closed at 174.277099609375 on 2024-03-04 00:00:00
Short position open at 187.1961212158203 on 2024-02-08 00:00:00
Short position closed at 174.277099609375 on 2024-03-04 00:00:00
Short position open at 186.27044677734375 on 2024-02-12 00:00:00
Short position closed at 174.277099609375 on 2024-03-04 00:00:00
Short position open at 184.17037963867188 on 2024-02-13 00:00:00
Short position closed at 174.277099609375 on 2024-03-04 00:00:00
Short position open at 183.28456115722656 on 2024-02-14 00:00:00
Short position closed at 169.32049560546875 on 2024-03-05 00:00:00
Short position open at 182.99594116210938 on 2024-02-15 00:00:00
Short position closed at 169.32049560546875 on 2024-03-05 00:00:00
Short position open at 181.4532012939453 on 2024-02-16 00:00:00
Short position closed at 183.50350952148438 on 2024-02-22 00:00:00
Short position open at 180.7067413330078 on 2024-02-20 00:00:00
Short position closed at 183.50350952148438 on 2024-02-22 00:00:00
Short position open at 181.46316528320312 on 2024-02-21 00:00:00
Short position closed at 183.50350952148438 on 2024-02-22 00:00:00
Short position open at 183.50350952148438 on 2024-02-22 00:00:00
Short position closed at 174.277099609375 on 2024-03-04 00:00:00
Short position open at 181.66224670410156 on 2024-02-23 00:00:00
Short position closed at 169.32049560546875 on 2024-03-05 00:00:00
Short position open at 180.30860900878906 on 2024-02-26 00:00:00
Short position closed at 181.77171325683594 on 2024-02-27 00:00:00
Short position open at 181.77171325683594 on 2024-02-27 00:00:00
Short position closed at 169.32049560546875 on 2024-03-05 00:00:00
Short position open at 180.56739807128906 on 2024-02-28 00:00:00
Short position closed at 169.32049560546875 on 2024-03-05 00:00:00
Short position open at 179.90054321289062 on 2024-02-29 00:00:00
Short position closed at 169.32049560546875 on 2024-03-05 00:00:00
Short position open at 178.81565856933594 on 2024-03-01 00:00:00
Short position closed at 169.32049560546875 on 2024-03-05 00:00:00
Long position open at 172.90357971191406 on 2024-03-18 00:00:00
Long position closed at 170.56460571289062 on 2024-03-21 00:00:00
Long position open at 175.25250244140625 on 2024-03-19 00:00:00
Long position closed at 170.56460571289062 on 2024-03-21 00:00:00
Long position open at 177.83030700683594 on 2024-03-20 00:00:00
Long position closed at 170.56460571289062 on 2024-03-21 00:00:00
Short position open at 168.8527069091797 on 2024-04-03 00:00:00
Short position closed at 174.2173614501953 on 2024-04-11 00:00:00
Short position open at 168.026611328125 on 2024-04-04 00:00:00
Short position closed at 168.8726043701172 on 2024-04-09 00:00:00
Short position open at 168.7830352783203 on 2024-04-05 00:00:00
Short position closed at 174.2173614501953 on 2024-04-11 00:00:00
Short position open at 167.65834045410156 on 2024-04-08 00:00:00
Short position closed at 168.8726043701172 on 2024-04-09 00:00:00
Short position open at 168.8726043701172 on 2024-04-09 00:00:00
Short position closed at 174.2173614501953 on 2024-04-11 00:00:00
Short position open at 166.99147033691406 on 2024-04-10 00:00:00
Short position closed at 174.2173614501953 on 2024-04-11 00:00:00
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Long position open at 174.2173614501953 on 2024-04-11 00:00:00
Long position closed at 171.87841796875 on 2024-04-15 00:00:00
Long position open at 175.72027587890625 on 2024-04-12 00:00:00
Long position closed at 171.87841796875 on 2024-04-15 00:00:00
Short position open at 164.2245635986328 on 2024-04-19 00:00:00
Short position closed at 165.0605926513672 on 2024-04-22 00:00:00
Short position open at 165.0605926513672 on 2024-04-22 00:00:00
Short position closed at 166.11561584472656 on 2024-04-23 00:00:00
Short position open at 166.11561584472656 on 2024-04-23 00:00:00
Short position closed at 168.2256622314453 on 2024-04-24 00:00:00
Short position open at 168.2256622314453 on 2024-04-24 00:00:00
Short position closed at 169.09158325195312 on 2024-04-25 00:00:00
Long position open at 182.51817321777344 on 2024-05-03 00:00:00
Long position closed at 180.8560333251953 on 2024-05-06 00:00:00
Long position open at 183.70260620117188 on 2024-05-09 00:00:00
Long position closed at 182.4368438720703 on 2024-05-10 00:00:00
Short position open at 192.4730987548828 on 2024-06-10 00:00:00
Short position closed at 206.4561004638672 on 2024-06-11 00:00:00
Long position open at 206.4561004638672 on 2024-06-11 00:00:00
Long position closed at 219.53216552734375 on 2024-07-02 00:00:00
Long position open at 212.35630798339844 on 2024-06-12 00:00:00
Long position closed at 208.97763061523438 on 2024-06-20 00:00:00
Long position open at 213.52236938476562 on 2024-06-13 00:00:00
Long position closed at 211.77822875976562 on 2024-06-14 00:00:00
Long position open at 215.94422912597656 on 2024-06-17 00:00:00
Long position closed at 213.5721893310547 on 2024-06-18 00:00:00
Long position open at 216.02395629882812 on 2024-07-01 00:00:00
Long position closed at 227.056884765625 on 2024-07-08 00:00:00
Long position open at 219.53216552734375 on 2024-07-02 00:00:00
Long position closed at 232.1995849609375 on 2024-07-10 00:00:00
Long position open at 220.80787658691406 on 2024-07-03 00:00:00
Long position closed at 232.1995849609375 on 2024-07-10 00:00:00
Long position open at 225.5818328857422 on 2024-07-05 00:00:00
Long position closed at 223.42906188964844 on 2024-07-18 00:00:00
Long position open at 227.056884765625 on 2024-07-08 00:00:00
Long position closed at 223.42906188964844 on 2024-07-18 00:00:00
Long position open at 227.9139862060547 on 2024-07-09 00:00:00
Long position closed at 223.42906188964844 on 2024-07-18 00:00:00
Short position open at 228.1133270263672 on 2024-07-17 00:00:00
Short position closed at 208.56903076171875 on 2024-08-05 00:00:00
Short position open at 223.42906188964844 on 2024-07-18 00:00:00
Short position closed at 208.56903076171875 on 2024-08-05 00:00:00
Short position open at 223.5586395263672 on 2024-07-19 00:00:00
Short position closed at 208.56903076171875 on 2024-08-05 00:00:00
Short position open at 223.20982360839844 on 2024-07-22 00:00:00
Short position closed at 208.56903076171875 on 2024-08-05 00:00:00
Short position open at 224.25628662109375 on 2024-07-23 00:00:00
Short position closed at 208.56903076171875 on 2024-08-05 00:00:00
Short position open at 218.06710815429688 on 2024-07-30 00:00:00
Short position closed at 221.3361053466797 on 2024-07-31 00:00:00
Short position open at 217.62855529785156 on 2024-08-01 00:00:00
Short position closed at 219.12355041503906 on 2024-08-02 00:00:00
Short position open at 219.12355041503906 on 2024-08-02 00:00:00
Short position closed at 206.5358428955078 on 2024-08-06 00:00:00
Short position open at 222.28079223632812 on 2024-09-03 00:00:00
Short position closed at 228.3673858642578 on 2024-09-19 00:00:00
```

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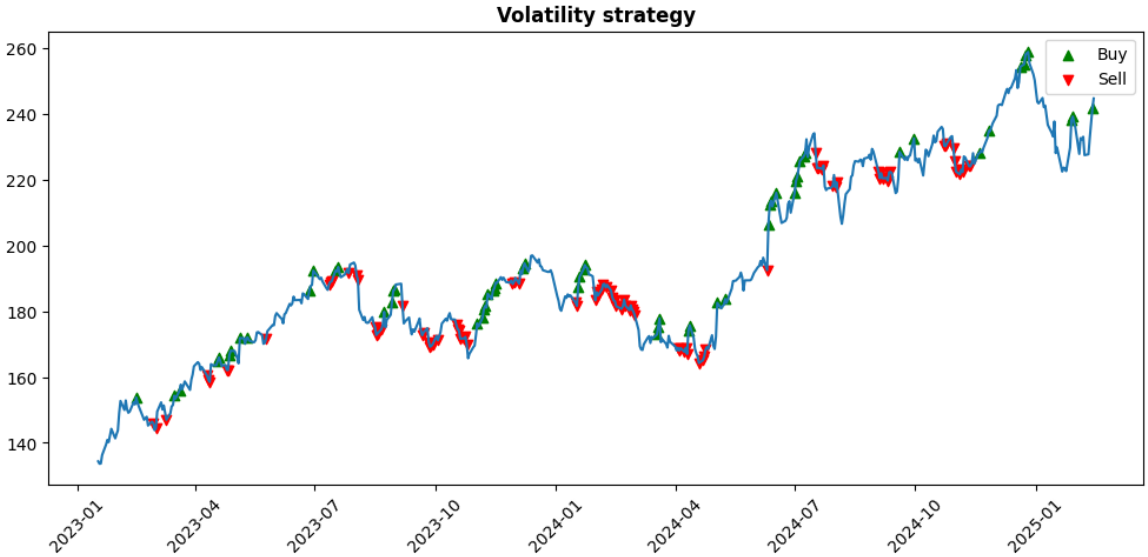
Short position open at 220.36500549316406 on 2024-09-04 00:00:00
Short position closed at 221.8916473388672 on 2024-09-05 00:00:00
Short position open at 221.8916473388672 on 2024-09-05 00:00:00
Short position closed at 228.3673858642578 on 2024-09-19 00:00:00
Short position open at 220.33506774902344 on 2024-09-06 00:00:00
Short position closed at 222.17103576660156 on 2024-09-11 00:00:00
Short position open at 220.4248809814453 on 2024-09-09 00:00:00
Short position closed at 222.17103576660156 on 2024-09-11 00:00:00
Short position open at 219.62661743164062 on 2024-09-10 00:00:00
Short position closed at 222.17103576660156 on 2024-09-11 00:00:00
Short position open at 222.17103576660156 on 2024-09-11 00:00:00
Short position closed at 228.3673858642578 on 2024-09-19 00:00:00
Short position open at 222.28079223632812 on 2024-09-12 00:00:00
Short position closed at 228.3673858642578 on 2024-09-19 00:00:00
Long position open at 228.3673858642578 on 2024-09-19 00:00:00
Long position closed at 225.97265625 on 2024-09-23 00:00:00
Long position open at 232.48831176757812 on 2024-09-30 00:00:00
Long position closed at 225.71322631835938 on 2024-10-01 00:00:00
Short position open at 230.25323486328125 on 2024-10-23 00:00:00
Short position closed at 232.88743591308594 on 2024-10-28 00:00:00
Short position open at 230.06365966796875 on 2024-10-24 00:00:00
Short position closed at 232.88743591308594 on 2024-10-28 00:00:00
Short position open at 230.9018096923828 on 2024-10-25 00:00:00
Short position closed at 232.88743591308594 on 2024-10-28 00:00:00
Short position open at 229.59469604492188 on 2024-10-30 00:00:00
Short position closed at 232.61424255371094 on 2024-11-25 00:00:00
Short position open at 225.4138946533203 on 2024-10-31 00:00:00
Short position closed at 226.98043823242188 on 2024-11-07 00:00:00
Short position open at 222.4204864501953 on 2024-11-01 00:00:00
Short position closed at 226.98043823242188 on 2024-11-07 00:00:00
Short position open at 221.52244567871094 on 2024-11-04 00:00:00
Short position closed at 222.95928955078125 on 2024-11-05 00:00:00
Short position open at 222.95928955078125 on 2024-11-05 00:00:00
Short position closed at 226.98043823242188 on 2024-11-07 00:00:00
Short position open at 222.23089599609375 on 2024-11-06 00:00:00
Short position closed at 226.98043823242188 on 2024-11-07 00:00:00
Short position open at 223.98373413085938 on 2024-11-11 00:00:00
Short position closed at 227.9693603515625 on 2024-11-14 00:00:00
Short position open at 223.98373413085938 on 2024-11-12 00:00:00
Short position closed at 227.9693603515625 on 2024-11-14 00:00:00
Long position open at 228.02928161621094 on 2024-11-19 00:00:00
Long position closed at 242.3834991455078 on 2024-12-03 00:00:00
Long position open at 234.8018341064453 on 2024-11-26 00:00:00
Long position closed at 247.4978790283203 on 2024-12-10 00:00:00
Long position open at 254.21051025390625 on 2024-12-20 00:00:00
Long position closed at 251.9230194091797 on 2024-12-30 00:00:00
Long position open at 254.98965454101562 on 2024-12-23 00:00:00
Long position closed at 251.9230194091797 on 2024-12-30 00:00:00
Long position open at 257.91644287109375 on 2024-12-24 00:00:00
Long position closed at 255.30929565429688 on 2024-12-27 00:00:00
Long position open at 258.7355041503906 on 2024-12-26 00:00:00
Long position closed at 255.30929565429688 on 2024-12-27 00:00:00
Long position open at 237.99832153320312 on 2025-01-28 00:00:00
Long position closed at 235.74081420898438 on 2025-01-31 00:00:00
Long position open at 239.0971221923828 on 2025-01-29 00:00:00
Long position closed at 237.3290557861328 on 2025-01-30 00:00:00
```

```
Long position open at 241.52999877929688 on 2025-02-13 00:00:00

--------------------------------
Number of Trades: 155
Wins: 35
Losses: 120
Total Returns: 17.42%
```



In [13]:
```
def backtest_strategy_bt(data: pd.DataFrame, signal_name: str, stra
    signal = data[signal_name].astype(float).to_frame().copy()
    signal.rename(columns = {signal_name:'Close'}, inplace = True)
    strategy = bt.Strategy(strategy_name, [bt.algos.WeighTarget(sig
    backtest = bt.Backtest(strategy, data)

    return backtest

data_bt = apple_stock_data
data_bt['atr_signal'] = atr_signal['signal']
backtest_atr_bt = backtest_strategy_bt(data_bt, 'atr_signal', "Vola

result = bt.run(backtest_atr_bt)

#Plot the backtest result
result.display()
result.plot(title='Backtest result: Volatility strategy')
```

```
100%|██████████| 1/1 [00:00<00:00,  9.14it/s]
Stat                 Volatility strategy
-------------------  ---------------------
Start                2023-01-02
End                  2025-02-14
Risk-free rate       0.00%

Total Return         -6.90%
Daily Sharpe         -0.20
Daily Sortino        -0.27
CAGR                 -3.32%
Max Drawdown         -19.09%
Calmar Ratio         -0.17
```

```
MTD                   1.27%
3m                    1.69%
6m                    -1.97%
YTD                   0.99%
1Y                    0.44%
3Y (ann.)             -3.32%
5Y (ann.)             -
10Y (ann.)            -
Since Incep. (ann.)   -3.32%

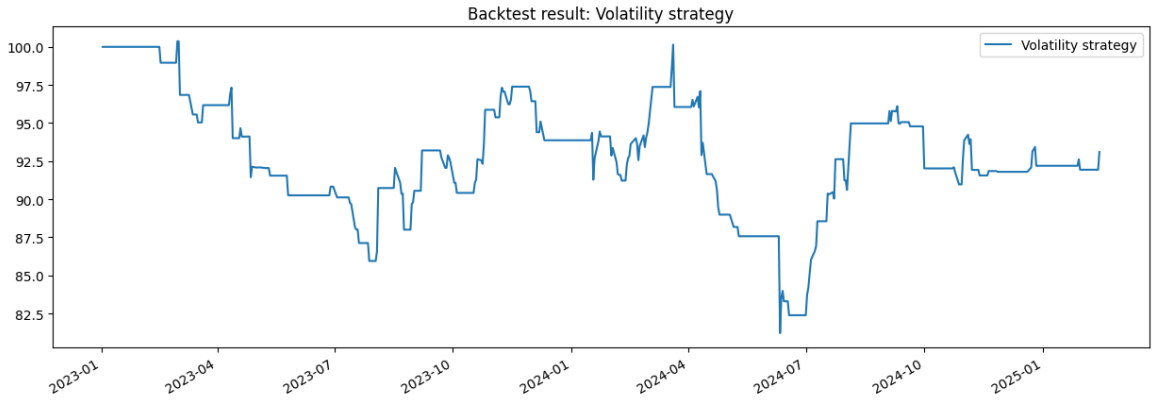
Daily Sharpe          -0.20
Daily Sortino         -0.27
Daily Mean (ann.)     -2.55%
Daily Vol (ann.)      12.89%
Daily Skew            -1.29
Daily Kurt            21.51
Best Day              4.82%
Worst Day             -7.27%

Monthly Sharpe        -0.20
Monthly Sortino       -0.35
Monthly Mean (ann.)   -2.60%
Monthly Vol (ann.)    13.21%
Monthly Skew          0.65
Monthly Kurt          1.75
Best Month            10.74%
Worst Month           -7.36%

Yearly Sharpe         -0.20
Yearly Sortino        -0.32
Yearly Mean           -0.40%
Yearly Vol            1.96%
Yearly Skew           -
Yearly Kurt           -
Best Year             0.99%
Worst Year            -1.78%

Avg. Drawdown         -10.07%
Avg. Drawdown Days    363.50
Avg. Up Month         2.66%
Avg. Down Month       -2.87%
Win Year %            50.00%
Win 12m %             20.00%
```

Out[13]: <Axes: title={'center': 'Backtest result: Volatility strategy'}>


Backtest result: Volatility strategy
Volatility strategy
100.0
97.5
95.0
92.5
90.0
87.5
85.0
82.5
2023-01
2023-04
2023-07
2023-10
2024-01
2024-04
2024-07
2024-10
2025-01